

# Mohamed Omar

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## EDUCATION

### University of Minnesota

*Bachelor of Computer Engineering*

Minneapolis, MN

*Expected: December 2027*

**GPA: 3.75 (4x Dean's List) & Relevant Coursework:** Embedded Systems Design, Robot Vision, Microcontrollers, Analog & Digital Electronics, Operating Systems, Computer & Machine Architecture, Algorithms & Data Structures

## EXPERIENCE

### Vehicle Controls Engineering Intern

June 2026 – August 2026

*Arc Boats*

*Torrance, CA*

- Designed a closed loop steering controller using Simulink for an electric boat VCU, replacing a one time rudder calibration with a gain-scheduled proportional control algorithm driven by speed, helm angle & IMU yaw rate, cutting hands off heading drift by 58% across 5 on water validation sessions.
- Resolved a field reported throttle mismatch fault, tracing it to common-mode voltage shifts from shared 5 V/ground harness impedance and correcting it in VCU firmware via L1 normalization of the ratiometric channel pair, preserving correct neutral/forward/reverse boundary behavior and improving tolerable harness resistance from 10 ohms to 20 ohms, deploying the fix fleet-wide via OTA.
- Automated a manual ballast calibration step of production acceptance testing with a Stateflow state machine driving all 5 ballast zones in parallel, cutting a 21 min technician procedure (23% of on water ATP time) to a 5–6 min automated run.
- Delivered 9 additional VCU tickets spanning supporting electrical HV charging improvements, thermal & OTA state handling, complete touchscreen/CAN commanded chair bring up, CAN fault diagnostics across firmware, DBC and calibration changes + more.

### Battery Systems Engineer

August 2025 – Present

*Solar Vehicle Team – University of Minnesota*

*Minneapolis, MN*

- Designed & implemented battery management systems for a high voltage (150V) lithium-ion pack, including writing STM32 firmware & designing PCBs (Altium Designer) for isoSPI cell monitoring, CAN communication & passive balancing.
- Developed embedded firmware (C/C++) for HV safety and power management in a race critical environment, implementing contactor sequencing, precharge state machines, ADC based current/voltage monitoring & fault handling logic.
- Performed hands-on battery pack assembly and validation, including PCB debugging, welding cells/soldering components and electrical testing using oscilloscopes, multimeters, and power supplies.

### Undergraduate Research Assistant

August 2025 – Present

*Computer Networking & AI Research Lab, University of Minnesota*

*Minneapolis, MN*

- Implemented and tested autonomous vehicle and teleoperation systems using multimodal sensor fusion (camera, LiDAR, radar) to support safe operation over long distances and bandwidth constrained networks.
- Developed ROS 2 based mobile robot autonomy systems using embedded edge computing on an NVIDIA Jetson.

### Software Engineering Intern

May 2025 – August 2025

*TalkRemit*

*London, United Kingdom*

- Implemented RESTful API features for a cross platform money transfer application, including a real time quote feature fetching live FX rates and fees that improved transfer speeds by 20%, alongside SSL encryption and Anti-Money Laundering safeguard improvements.

## PROJECTS - [OMARSLAB.COM](https://OmarsLab.com)

### Baby Monitor | PIC24, C, ADC Interrupts, Altium Designer, Fusion 360

- Developed a dual PIC24 microcontroller system with real-time ADC interrupts driving sensor and audio processing, PWM LED control, and LCD output, then independently scaled it into a manufacturable custom PCB and enclosure in Altium Designer and Fusion 360

### AllerSight | O1 Hackathon - Raspberry Pi, Computer Vision, Python

- Owned the embedded hardware and computer vision subsystem of a real-time allergen safety system, prototyping Raspberry Pi smart glasses that streamed video to a backend for ingredient detection across hands, tools, and surfaces, maintaining a contamination state that reset on detected wash events

### Autonomous Obstacle Avoiding RC Car | ESP32, Embedded Systems, C/C++, PWM Motor Control

- Built an autonomous RC car on an ESP32 with a servo-mounted VL53L1X time of flight sensor, implementing closed-loop PWM motor control and interrupt driven distance sensing for obstacle detection and path correction

## TECHNICAL SKILLS

**Languages & Tools:** C / C++, Simulink / MATLAB, TSMaster, PCB Design (Altium Designer), ROS 2, Python, Git

**Embedded & Protocols:** STM32, ESP32, PIC24, Raspberry Pi, CAN Bus, SPI, I2C, isoSPI, RTOS, ADC, PWM, DBC, OTA